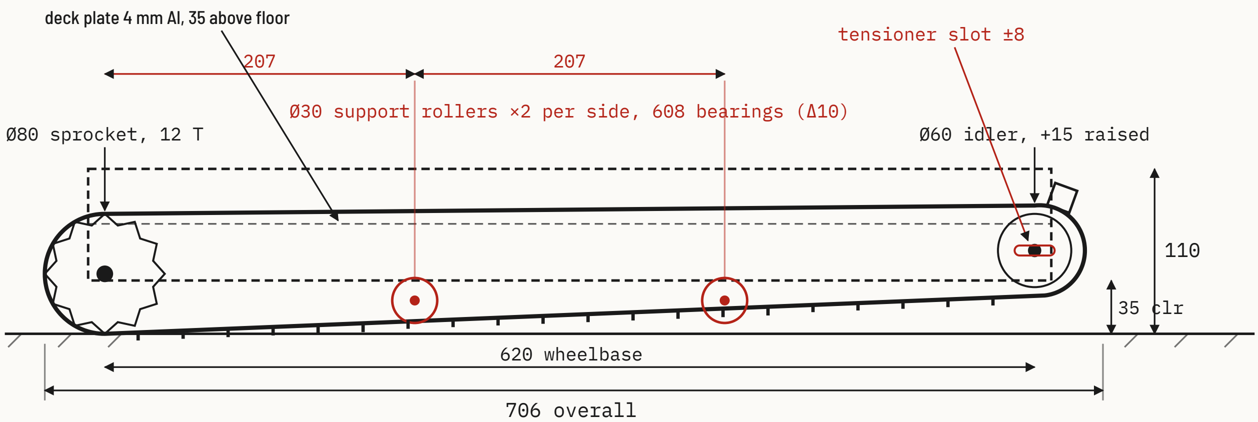
A B C D E F G H

GENERAL ARRANGEMENT - SIDE, LEVEL GROUND



scale 0.9 px/mm · rear ← → front · 49 links × 30 pitch = 1470 track

Δ10 wheelbase. The 620 mm bottom run needs two support rollers per side or it sags and the lugs skip the sprocket under climb torque. Track loop length is set by the link count so the idler slot only takes up wear.

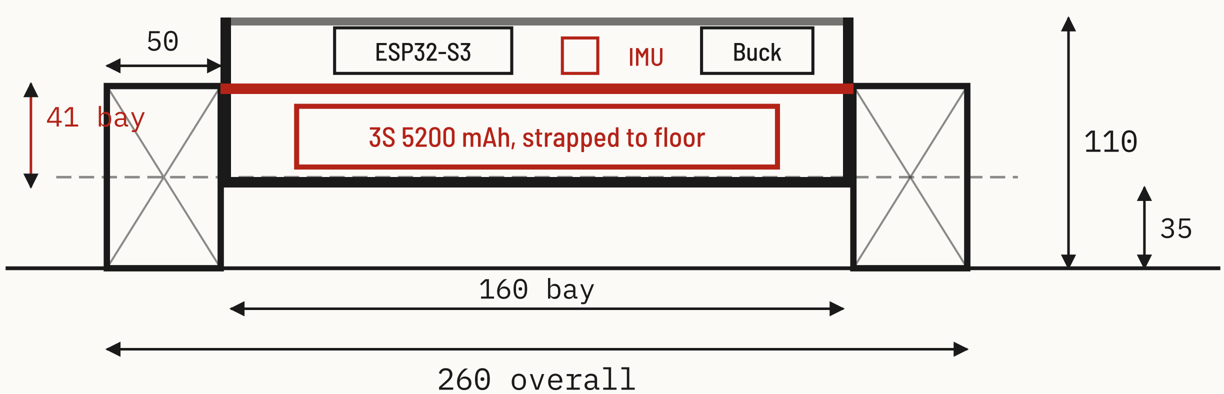
1 1

SECTION A-A - THROUGH THE BATTERY BAY

Side plates 4 mm 6061 Al · deck and floor 4 mm · lid 2 mm

Axle Ø8 through 608 bearings in the side plates; motors inboard

Mass split: 1.35 kg below deck, 0.6 kg above, 2.05 kg tracks, rollers and frame

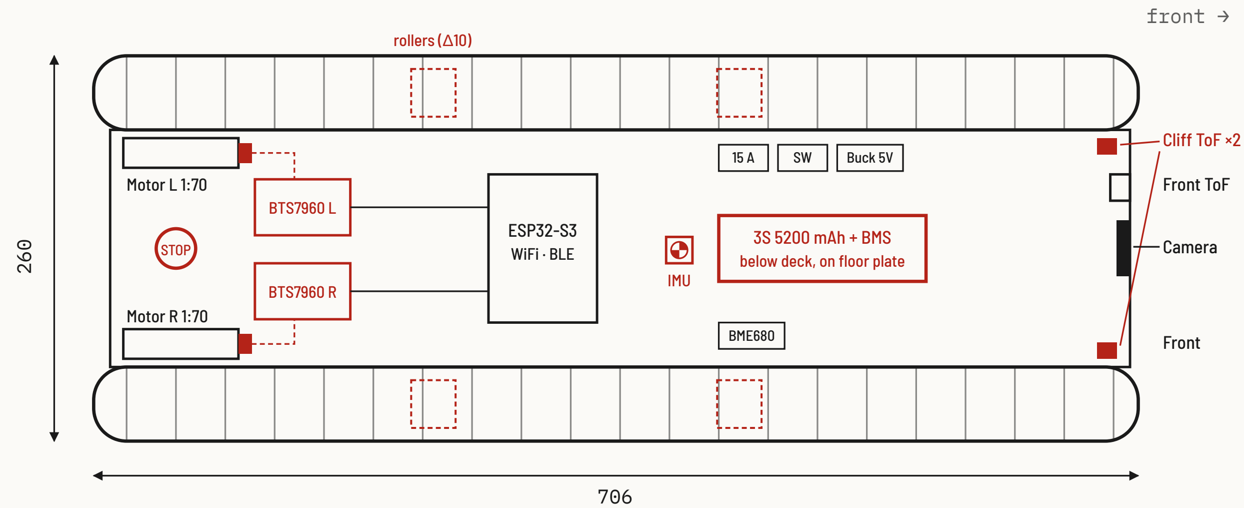


2 2

Everything heavy lives under the deck plate, on the floor, which is what puts the CG at 88 mm on a 110 mm tall vehicle. The section is unchanged by Δ10; only the length grew. The IMU is bonded to the deck directly above the battery, not to the lid.

PLAN VIEW - INTERNAL LAYOUT

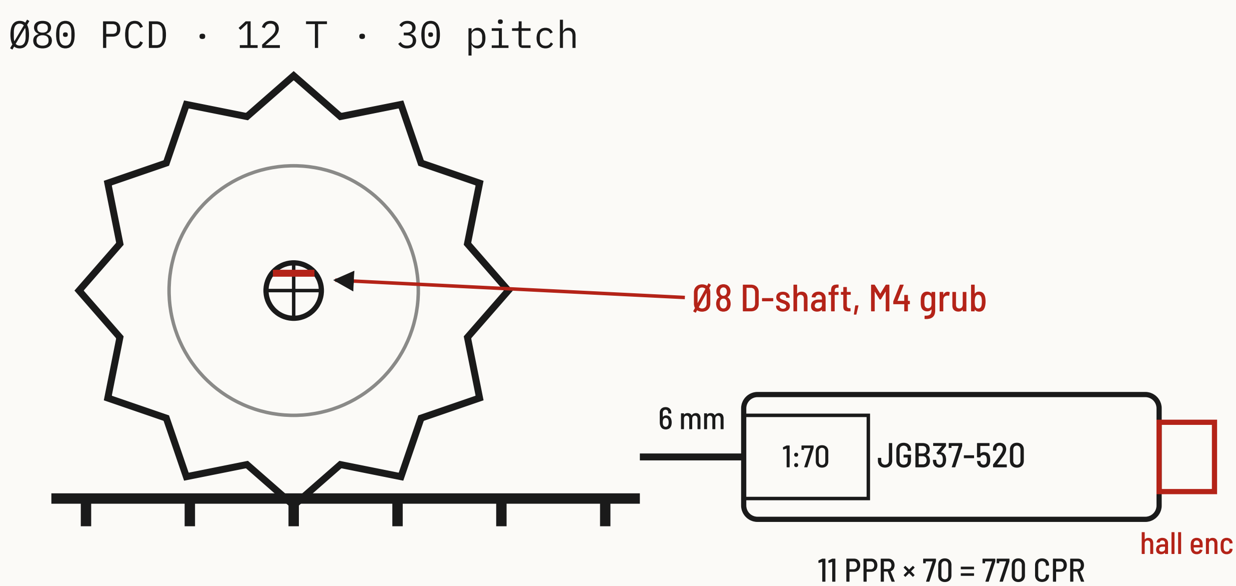
3

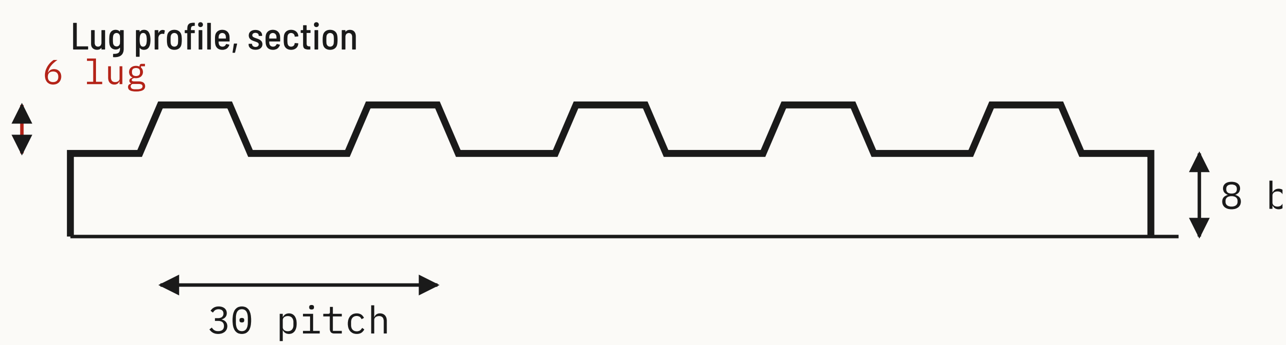


Heavy items sit forward and on the floor plate. The IMU is mounted at the CG so pitch reads clean of chassis flex. The I^{2}C chain to the front rangefinders is now about 650 mm, so the pull-ups sit at the far end. Encoder returns (dashed) close the speed loop per side.

DRIVE MODULE AND TRACK LUG

4





TPU 95A or moulded rubber, chevron lug, 50 wide

edge lug undercut 10° for nosing grip

5 One bolt-on module per side: motor, encoder, axle, bearings and sprocket. The lug pitch equals the sprocket pitch so the drive is positive, not friction. Δ10 adds two Ø30 support rollers per side on M8 stub axles between sprocket and idler. 5

REVISIONS 1.0 → 2.2

MARK	CHANGE AND BASIS
Δ1	**Track length 210 → 450 mm.** Sized from nosing pitch 333 mm × 1.35. Raised front idler for 35° approach.
Δ2	**Battery moved low and forward.** CG 150 mm from rear axle, 88 mm high. ~~Rear battery + camera mast~~ removed.
Δ3	**Camera fixed, 20° down, on front face.** ~~Pan/tilt servos~~ deleted; no mast, no servo power.
Δ4	**Drive: 2 × BTS7960, one per side.** ~~L298N~~ removed. Motors: 2 × 37 mm 12 V gearmotor, 1:70, 100 rpm, hall encoders.
Δ5	**Power tree completed.** 3S 5200 mAh with BMS, 15 A blade fuse, main switch, XT60, 5 V 3 A buck. Motors fed direct from battery.
Δ6	**One controller.** ESP32-S3 with on-board OV2640. ~~Second ESP32-CAM~~ and its undefined link removed.
Δ7	**Sensors all 3.3 V.** BME680 only (~~MQ-135~~ dropped), VL53L1X front, 2 × VL53L0X cliff sensors under the front idler, MPU-6050 IMU at the CG.
Δ8	**Descent is controlled.** IMU pitch selects descent mode: 40% speed cap, active PWM braking, hold on stick release, cliff stop.
Δ9	**Safety path.** BLE gamepad is the primary link, WiFi is telemetry only. 500 ms link watchdog stops the motors. Physical E-stop opens the motor supply.
Δ10	**Wheelbase 384 → 620 mm, overall 470 → 706 mm.** Rigid-body simulation of Rev 2.1 showed the rover cannot climb the first 180 mm riser from the floor: with the rear on the floor and the nosing sliding along the belt, the CG never passes the step edge and the rover backflips at any friction, torque or speed. At 620 mm the front reaches the second nosing as the rear lifts. Adds two Ø30 support rollers per side, 49-link tracks, longer plates and lid. Mass 3.07 → 4.0 kg.

REV	DATE	DESCRIPTION
1.0	2025-05-18	Concept sheet
2.0	2026-09-02	Audit fixes Δ1-Δ9, geometry to scale
2.1	2026-09-02	Detail issue: GA, section, drive, electrical, BOM
2.2	2026-09-02	Δ10: wheelbase 620, support rollers, 49-link tracks, 4.0 kg

PROJECT
RoboTank

TITLE
Stair-climbing tracked IoT rover
General arrangement

DRAWING NO.	SHEET	REV	SCALE
RT-100-01	1 / 4	2.2	0.9 px/mm
UNITS	**PROJECTION**	**DATE**	**STATUS**
mm	Third angle	2026-09-02	For review
CHECKED	**APPROVED**		
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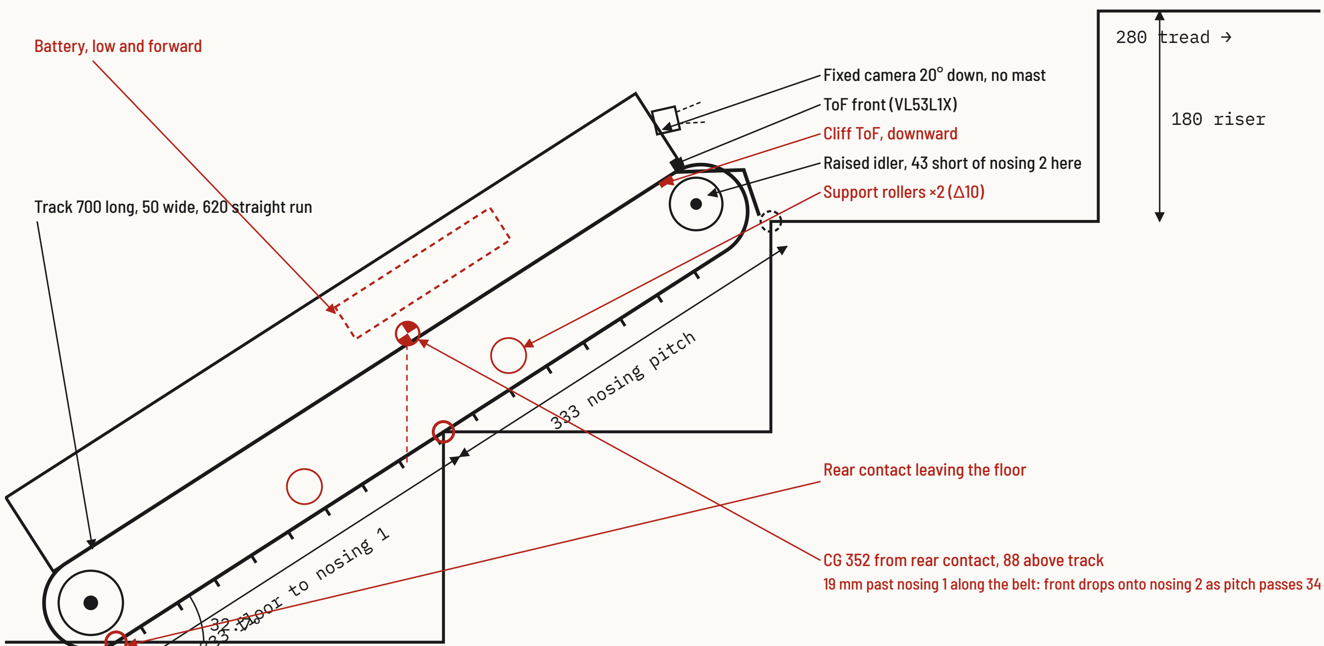
SIDE ELEVATION - ON STAIR, TO SCALE



scale 0.8 px per mm · overall 706 L × 260 W × 110 H · pose from the rigid-body simulation at t = 2.9 s

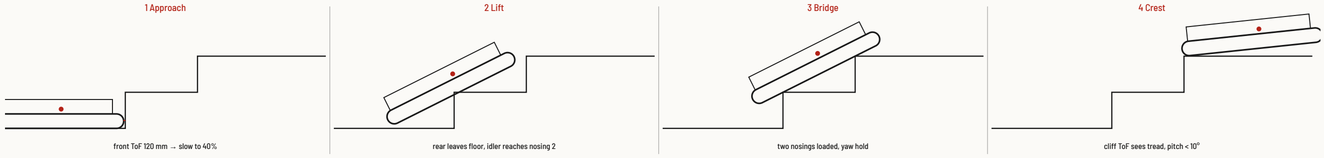
The first riser is the hard case. From the floor, the belt line through nosing 1 needs 663 mm to reach nosing 2. The 620 mm straight run leaves the raised idler 43 mm short at the instant the rear lifts, but the CG is already 19 mm past nosing 1 along the belt, so the rover rotates forward, not back, and the idler lands on nosing 2 within a degree. Rev 2.1 at 450 mm had its CG 100 mm behind the pivot at this moment and went over backwards.

SIZING CHECK

ITEM	V2.1	V2.2 Δ10	BASIS
Track length	450 mm	700 mm	620 straight run + idler reach vs 663 floor to nosing 2
First riser from floor	backflip	climbs	Rigid-body simulation, friction 1.0
Step / pitch rating	180 mm, 35°	180 mm, 35°	Standard 180 × 280 stair
Mass, all-up	3.07 kg	4.0 kg	BOM sheet 4
Tractive force at 35°	26 N	31 N	m·g·sin θ + 8 N rolling
Torque per side	0.52 N·m	0.61 N·m	15.5 N × 40 mm sprocket
Motor rating	1.0 N·m, 1.9×	1.0 N·m, 1.6×	37 mm 1:70 gearmotor
Climb speed	0.29 m/s	0.29 m/s	70% cap, 40 mm sprocket
Motor driver	2 × BTS7960	2 × BTS7960	43 A, 16 mΩ
Peak current	6 A	7 A	2 × 3 A climb + 1 A logic
Runtime	~1.5 h	~1.3 h	5.2 Ah at 3.5 A average
Support rollers	none	2 per side	620 mm run sags 5 to 8 mm unsupported
Control latency	< 30 ms BLE	< 30 ms BLE	Cloud round trip unfit for stairs

CLIMB SEQUENCE AND SENSOR CUES

5 5



The operator only holds forward. Speed caps, mode changes and the yaw hold come from the front ToF, the IMU and the encoders in that order. With the Δ10 length the lift phase ends on the second nosing instead of a pivot; on descent the sequence runs in reverse and the cliff sensors are the first cue.

6 6

REV	DATE	DESCRIPTION
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2.1	2026-09-02	Detail issue: GA, section, drive, electrical, BOM
2.2	2026-09-02	Δ10: wheelbase 620, support rollers, 49-link tracks, 4.0 kg

PROJECT
RoboTank

TITLE
Stair-climbing tracked IoT rover
Stair performance

DRAWING NO.	SHEET	REV	SCALE
RT-100-02	2 / 4	2.2	0.8 px/mm
UNITS	PROJECTION	DATE	STATUS
mm	Third angle	2026-09-02	For review
CHECKED	APPROVED		
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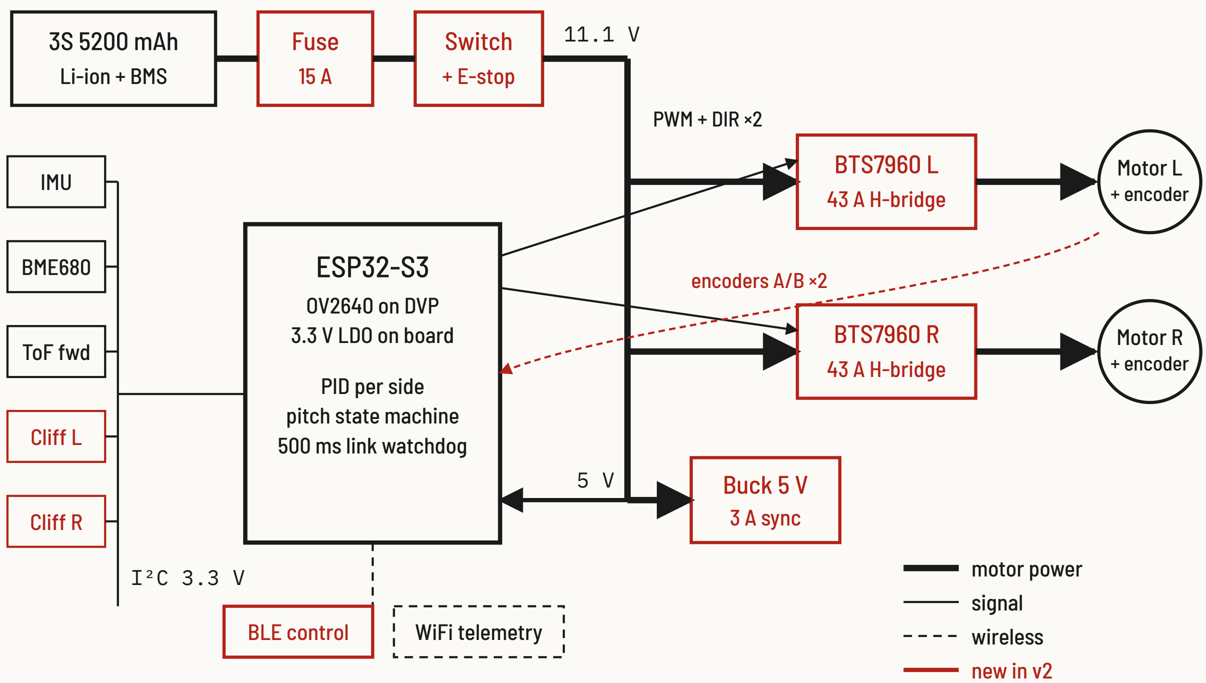
POWER AND SIGNAL TREE



Motor current never passes through the logic supply. The HC-SR04 and MQ-135 are gone, so nothing on the bus needs 5 V or a level shifter. Control arrives over BLE with a 500 ms watchdog; the cloud only ever receives telemetry.

ESP32-S3 PIN MAP

GPIO	SIGNAL	TO
1	I^2C SDA	IMU, BME680, ToF ×3
2	I^2C SCL	same bus, 3.3 V, 4k7
14	PWM L	BTS7960 L RPWM/LPWM via DIR
21	DIR L	BTS7960 L
47	PWM R	BTS7960 R
48	DIR R	BTS7960 R
19 / 20	ENC L A/B	PCNT unit 0
45 / 46	ENC R A/B	PCNT unit 1
3	VBAT sense	ADC, 10k:2k2 divider
0	E-stop sense	NC contact, pull-up
4-18, 38-42	DVP camera	OV2640, fixed by board

Pins chosen from those the WROOM-CAM board leaves free of the camera bus. GPIO 0, 3, 45 and 46 are strapping pins: keep them high-impedance at boot and verify on the exact board revision.

HARNESS AND CONNECTORS

4

RUN	WIRE	CONNECTOR	NOTE
Battery → fuse → switch → bus	14 AWG	XT60	Under 120 mm total, E-stop NC in series
Bus → BTS7960 ×2	16 AWG	Screw terminal	1000 µF low-ESR at each driver
BTS7960 → motor	16 AWG	XT30	Twisted pair, ferrite bead
Bus → buck → ESP32 5 V	22 AWG	JST-XH 2	Ground star at buck output
ESP32 → BTS7960 logic	26 AWG	JST-XH 4	PWM, DIR, EN, GND per driver
Encoder → ESP32	26 AWG	JST-XH 6	Shielded, 3.3 V hall supply
I²C bus	26 AWG	JST-SH 4 (Qwiic)	Daisy chain, about 650 mm total with Δ10; 4k7 pull-ups at the front end
Cliff ToF ×2	26 AWG	JST-SH 4 + XSHUT	Separate XSHUT lines for addressing

5

Motor current and logic share only the battery negative at one star point at the buck. Signal runs never lie parallel to motor leads.

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PROJECT
RoboTank

TITLE
Stair-climbing tracked IoT rover
Electrical

6

DRAWING NO.	SHEET	REV	SCALE
RT-100-03	3 / 4	2.2	NTS
UNITS	**PROJECTION**	**DATE**	**STATUS**
mm	Third angle	2026-09-02	For review
CHECKED	**APPROVED**		
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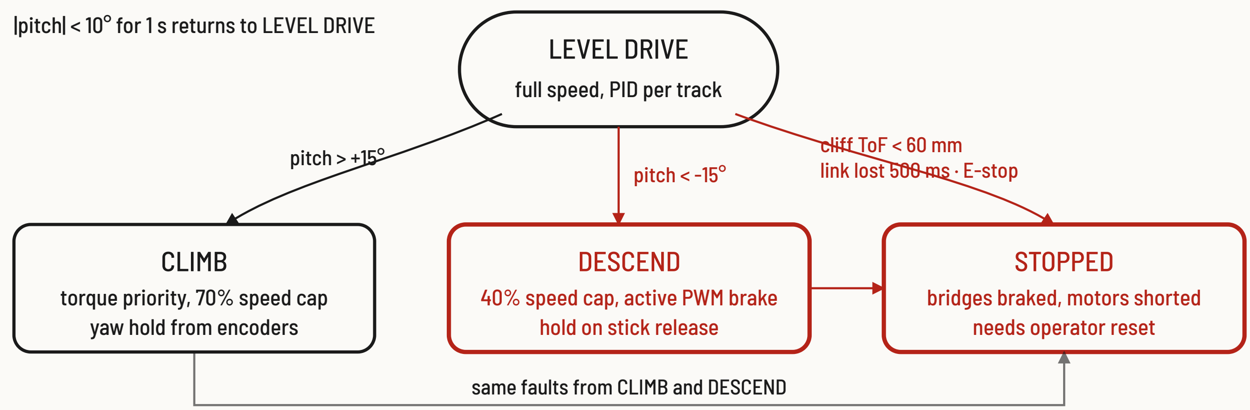
STAIR CONTROL STATES



Pitch from the IMU, not the operator, decides the mode. Descent exists as its own state so the speed cap and braking cannot be forgotten by a remote driver. Every fault path ends in the same braked, latched stop.

FIRMWARE LOOPS

LOOP	RATE	DOES	FAIL ACTION
Encoder count	HW PCNT	Quadrature on 4 pins, no ISR load	-
Speed PID	200 Hz	Per-track velocity loop, yaw hold from L-R count delta	PWM 0, brake
IMU	100 Hz	Complementary filter, pitch and roll, mode state machine	Hold last mode, warn
ToF front and cliff	20 Hz	Front slow-down at 120 mm, cliff stop at 60 mm	STOPPED
Link watchdog	10 Hz	BLE packet age; stop after 500 ms silence	STOPPED
Battery	1 Hz	Pack volts via divider; cut motors at 9.6 V	Return to level, stop
Telemetry	10 Hz	WiFi MQTT: pitch, speeds, current, BME680, mode	Drop, never block
Camera stream	15 fps	OV2640 VGA MJPEG on a separate core	Drop frames

Control loops run on core 1, camera and WiFi on core 0. Nothing on core 1 waits on the network.

BILL OF MATERIALS

#	ITEM	SPEC	QTY	MASS G
1	Controller	ESP32-S3-WROOM-1 N8R8 CAM board, OV2640 120° lens	1	25
2	Gearmotor	JGB37-520, 12 V, 1:70, 100 rpm, 1.0 N·m, hall encoder	2	400
3	Motor driver	BTS7960 43 A module, heatsinked	2	80
4	Battery	3S 5200 mAh Li-ion 18650 ×6, 20 A BMS, XT60	1	330
5	Protection	15 A blade fuse + holder, 20 A rocker, Ø22 NC mushroom E-stop	1 set	60
6	Logic supply	5 V 3 A synchronous buck, 470 µF at output	1	15
7	IMU	MPU-6050 breakout, deck mounted	1	3

4

8	Environment	BME680 breakout, rear vent	1	3
9	Range	VL53L1X front; VL53L0X cliff, XSHUT addressed	1 + 2	6
10	Drive train	Ø80 12 T sprocket, Ø60 idler, Ø8 axles, Ø30 support rollers ×4 on M8 stubs, 608 bearings ×16	2 sets	440
11	Tracks	49 links × 30 mm pitch, 50 wide, TPU 95A lugs	2	770
12	Frame	4 mm 6061 side 706 × 110, floor and deck 640 × 160; 2 mm lid; M4 hardware	1	1750
13	Harness	16 AWG motor, 22 AWG signal, JST-XH, XT60	1	120
14	Controller link	BLE gamepad (Xbox or 8BitDo), HID mode	1	-
	Total all-up			**4002**

Sizing sheet 2 uses a 4.0 kg budget. Δ10 added 930 g, mostly plates and links. Frame mass is the lever if it grows: 3 mm side plates save about 350 g.

5

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PROJECT
RoboTank

TITLE
Stair-climbing tracked IoT rover
Control, firmware and BOM

6

DRAWING NO.	SHEET	REV	SCALE
RT-100-04	4 / 4	2.2	NTS
UNITS	**PROJECTION**	**DATE**	**STATUS**
mm	Third angle	2026-09-02	For review
CHECKED	**APPROVED**		
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